



**CALICUT UNIVERSITY – FOUR-YEAR UNDER  
GRADUATE PROGRAMME (CU-FYUGP)**

**BSc PHYSICS HONOURS**

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                     |                  |                   |                    |             |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Programme      | <b>B.Sc. Physics Honours</b>                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                   |                    |             |
| Course Title   | <b>PHOTONICS</b>                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                   |                    |             |
| Type of Course | <b>Major Elective (SPECIALIZATION II: PHOTONICS)</b>                                                                                                                                                                                                                                                                                                                                                                                |                  |                   |                    |             |
| Semester       | <b>V</b>                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |                   |                    |             |
| Academic Level | <b>300 → 399</b>                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                   |                    |             |
| Course Details | Credit                                                                                                                                                                                                                                                                                                                                                                                                                              | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 4                                                                                                                                                                                                                                                                                                                                                                                                                                   | 4                | -                 | -                  | 60          |
| Pre-requisites | Fundamental knowledge in Optics                                                                                                                                                                                                                                                                                                                                                                                                     |                  |                   |                    |             |
| Course Summary | Photonics is the science and technology of generating, controlling, and detecting photons, which are particles of light. This course covers topics such as the fundamentals of light-matter interaction, optical components and systems, laser technology and fiber optics. It's a multidisciplinary field that combines elements of physics, optics and materials science to harness light for a wide range of practical purposes. |                  |                   |                    |             |

**Course Outcomes (CO):**

| CO  | CO Statement                                                                                                                                            | Cognitive Level* | Knowledge Category # | Evaluation Tools used       |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------|-----------------------------|
| CO1 | Understand the concept and principles of energy levels, spontaneous emission and stimulated emission, optical gain, and threshold condition for lasing. | U                | C                    | Written exams and quizzes   |
| CO2 | Understand the principles and working of various laser systems.                                                                                         | An               | p                    | Presentations, written exam |
| CO3 | Giving a rigorous theoretical background and framework for a                                                                                            | U                | C                    | Written exams, Assignments  |

|                                                                                                                                                                                                    |                                                                                                                                                     |        |   |                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|----------------------------------------|
|                                                                                                                                                                                                    | nonlinear optical effect, followed by details of how such an effect is implemented in real applications.                                            |        |   |                                        |
| CO4                                                                                                                                                                                                | Understand the physical principles of optical fiber and the loss mechanisms in optical fiber. Demonstrate the understanding of fiber optic sensors. | U & Ap | C | Written exams and quizzes, experiments |
| CO5                                                                                                                                                                                                | Apply Photonics principles to real-world applications such as lasers and Optical fiber                                                              | U & Ap | C | Simple projects                        |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge (F) Conceptual Knowledge (C) Procedural Knowledge (P)<br>Metacognitive Knowledge (M) |                                                                                                                                                     |        |   |                                        |

**Detailed Syllabus:**

| Module    | Unit                                  | Content                                                                                                         | Hrs (48+12) | Marks (70) |
|-----------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------|------------|
| <b>I</b>  | <b>LASERS: AN INTRODUCTION</b>        |                                                                                                                 | <b>15</b>   | <b>25</b>  |
|           | 1                                     | Introduction, spontaneous and stimulated emission, main components of the laser                                 | 2           |            |
|           | 2                                     | Understanding optical amplification: The EDFA                                                                   | 1           |            |
|           | 3                                     | The resonator, The lasing action. Optical resonators                                                            | 3           |            |
|           | 4                                     | Einstein's equation and Conditions for light amplification, Metastable state, Population Inversion              | 3           |            |
|           | 5                                     | Cavity life time, The threshold condition                                                                       | 2           |            |
|           | 6                                     | Line shape function, Monochromaticity of laser beam                                                             | 2           |            |
|           | 7                                     | Laser Pumping – Two level system, Three level system                                                            | 2           |            |
|           |                                       | Sections 26.1, 26.1.1, 26.1.2, 26.1.3, 26.1.4, 26.1.5, 26.5, 26.6, 26.6.1, 26.6.2, 26.6.3, 26.7, 26.9 of Book 1 |             |            |
| <b>II</b> | <b>LASER SYSTEMS AND APPLICATIONS</b> |                                                                                                                 | <b>8</b>    | <b>10</b>  |
|           | 8                                     | Solid state lasers- Ruby Laser, Nd: YAG Laser                                                                   | 2           |            |
|           | 9                                     | Liquid Lasers – Dye lasers                                                                                      | 1           |            |
|           | 10                                    | Gas Lasers – Helium-Neon laser, CO <sub>2</sub> laser                                                           | 1           |            |
|           | 11                                    | Semiconductor Laser-Double heterojunction laser                                                                 | 2           |            |
|           | 12                                    | Chemical Laser – HCl laser, HF laser, Free Electron laser                                                       | 2           |            |

|            |                                                       |                                                                                                                                                                                                                                  |           |           |
|------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
|            |                                                       | Sections 7.1, 7.5, 10.1, 10.2, 8.1, 8.5.1, 9.1, 10.3, 10.3.1, 10.3.2 of Book-2                                                                                                                                                   |           |           |
| <b>III</b> | <b>NONLINEAR OPTICS</b>                               |                                                                                                                                                                                                                                  | <b>11</b> | <b>15</b> |
|            | 13                                                    | Harmonic generation, Second Harmonic generation, Phase Matching                                                                                                                                                                  | 3         |           |
|            | 14                                                    | Third Harmonic generation, Optical Mixing, Parametric generation of light                                                                                                                                                        | 3         |           |
|            | 15                                                    | Frequency Upconversion, Self-focusing of light                                                                                                                                                                                   | 2         |           |
|            | 16                                                    | Multiphoton processes- Two photon and three photon processes                                                                                                                                                                     | 3         |           |
|            |                                                       | Sections 13.1, 13.2, 13.3, 13.4, 13.5, 13.6, 13.7, 14.12, 14.2, 14.3, 14.7 of Book-2                                                                                                                                             |           |           |
| <b>IV</b>  | <b>OPTICAL FIBER BASICS</b>                           |                                                                                                                                                                                                                                  | <b>14</b> | <b>20</b> |
|            | 17                                                    | Introduction ,Some Historical Remarks                                                                                                                                                                                            | 1         |           |
|            | 18                                                    | Total Internal Reflection, The Numerical Aperture                                                                                                                                                                                | 2         |           |
|            | 19                                                    | Attenuation in Optical Fibers                                                                                                                                                                                                    | 2         |           |
|            | 20                                                    | Multimode Fibers, Pulse Dispersion in Multimode Optical Fibers, Dispersion and Maximum Bit Rates                                                                                                                                 | 4         |           |
|            | 21                                                    | Fiber Optic Sensors                                                                                                                                                                                                              | 2         |           |
|            | 22                                                    | TE Modes of a Symmetric Step Index Planar Waveguide (qualitative idea only) Physical Understanding of Modes, TM Modes of a Symmetric Step Index Planar Waveguide (qualitative idea only)                                         | 3         |           |
|            |                                                       | Sections 27.1, 27.2, 27.3, 27.4, 27.7, 27.8, 27.10, 27.11, 27.14, 28.2, 28.3, 28.4 of Book-1                                                                                                                                     |           |           |
| <b>V</b>   | <b>OPEN-ENDED MODULE: PROJECT / HANDS-ON TRAINING</b> |                                                                                                                                                                                                                                  | <b>12</b> |           |
|            |                                                       | Study the refraction of a laser beam in a glass slab and measure its refractive index using total internal reflection.                                                                                                           |           |           |
|            |                                                       | Determine the numerical aperture and acceptance angle of an optical fibre.                                                                                                                                                       |           |           |
|            |                                                       | Measure the divergence of an edge emitting diode laser beam by measuring the dimensions of the beam projected onto a screen at different distances. Hence to calculate the beam divergence and spot size of the given laser beam |           |           |

|  |                                                                                                                                       |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------|--|--|
|  | <b>References:</b><br>1. Optics by Ajoy Ghatak 5 <sup>th</sup> Edition (Book 1)<br>2. Laser and Nonlinear Optics by B B Laud (Book 2) |  |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------|--|--|

**Mapping of COs with PSOs and POs :**

|      | PSO 1 | PSO 2 | PSO 3 | PSO4 | PS O5 | PS O6 | PO1 | PO2 | PO3 | PO4 | PO5 | PO 6 | PO 7 |
|------|-------|-------|-------|------|-------|-------|-----|-----|-----|-----|-----|------|------|
| CO 1 | 3     | 0     | 0     | 0    | 0     | 0     | 3   | 2   | 0   | 0   | 0   | 0    | 1    |
| CO 2 | 3     | 2     | 0     | 0    | 0     | 0     | 3   | 2   | 0   | 0   | 0   | 0    | 1    |
| CO 3 | 3     | 0     | 0     | 0    | 0     | 0     | 3   | 2   | 0   | 0   | 0   | 0    | 2    |
| CO 4 | 3     | 0     | 0     | 3    | 0     | 0     | 3   | 2   | 0   | 2   | 0   | 0    | 3    |
| CO 5 | 3     | 0     | 0     | 3    | 0     | 0     | 3   | 2   | 0   | 2   | 0   | 0    | 2    |
| CO 6 | 3     | 0     | 0     | 0    | 0     | 0     | 3   | 2   | 0   | 0   | 0   | 0    | 1    |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| 0     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Discussion / Seminar
- InternalTheory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

**Mapping of COs to Assessment Rubrics**

|      | Internal Theory/<br>Practical Exam | Assignment<br>/Viva | Practical Skill<br>Evaluation | End Semester<br>Examinations |
|------|------------------------------------|---------------------|-------------------------------|------------------------------|
| CO 1 | ✓                                  | ✓                   |                               | ✓                            |
| CO 2 | ✓                                  | ✓                   |                               | ✓                            |
| CO 3 | ✓                                  | ✓                   |                               | ✓                            |
| CO 4 | ✓                                  | ✓                   |                               | ✓                            |
| CO 5 | ✓                                  | ✓                   |                               | ✓                            |
| CO 6 |                                    | ✓                   | ✓                             |                              |